

Problem: What is wrong with the following calculation?

$$\int_{-1}^3 \frac{1}{x^2} dx = \left. \frac{x^{-1}}{-1} \right|_{-1}^3 = -\frac{1}{3} - 1 = -\frac{4}{3}.$$

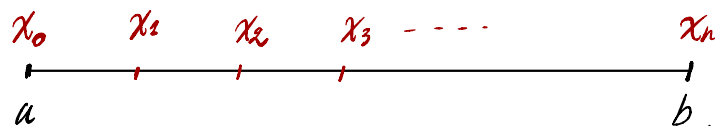
The Definite integral.

We know that using the limit

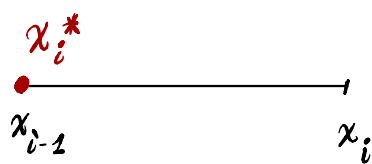
$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x.$$

we can compute the area under the curve $y=f(x)$.

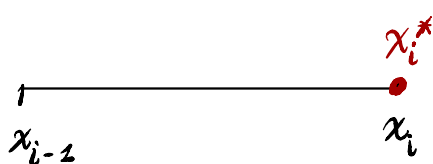
Def. If f is a function defined for $a \leq x \leq b$, we divide the interval $[a, b]$ into n subintervals of equal width $\Delta x = \frac{b-a}{n}$



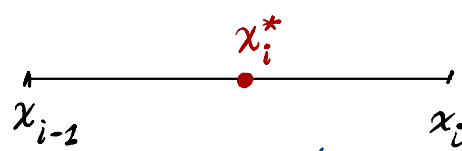
we denote $x_1^*, x_2^*, \dots, x_n^*$ by any sample points, x_i^* 's lies in the i -th subinterval.



lower Area.



upper Area.



Riemann Area.

Then, the definite integral of f from a to b .

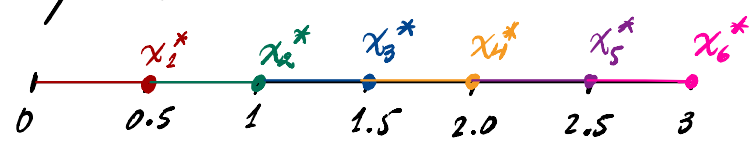
$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x.$$

provided that this limit and gives the same value for all possible choices of sample points.

Example. Evaluate the Riemann sum for $f(x) = x^3 - 6x$, $0 \leq x \leq 3$ with $n=6$ subintervals and taking the sample points endpoints

to be right endpoints.

Sln.


$$\Delta x = \frac{3-0}{6} = 0.5.$$

then

$$\begin{aligned} R_6 &= \sum_{i=1}^6 f(x_i^*) \Delta x \\ &= \Delta x \cdot \sum_{i=1}^6 f(x_i^*) \\ &= \frac{1}{2} \{ f(0.5) + f(1) + f(1.5) + f(2) + f(2.5) + f(3) \} \\ &= \frac{1}{2} \{ -2.875 - 5 - 5.625 - 4 + 0.625 + 9 \} = -3.9375. \end{aligned}$$

Thm 1. If f is a continuous function on $[a, b]$ or f has only finite number of jump discontinuities then f is integrable on $[a, b]$; i.e., the definite integral $\int_a^b f(x) dx$ exists.

Thm 2. If f is an integrable function on $[a, b]$ then

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x.$$

where $\Delta x = \frac{b-a}{n}$; $x_i = a + i \Delta x$

Example. Express

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n (x_i^3 + x_i \sin x_i) \Delta x$$

as an integral on the interval $[0, \pi]$.

Sln. notice that

$$\int_0^{\pi} f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x = \lim_{n \rightarrow \infty} \sum_{i=1}^n (x_i^3 + x_i \sin x_i) \Delta x.$$

then $f(x_i) = x_i^3 + x_i \sin x_i$ i.e., $f(x) = x^3 + x \sin x$.

So,

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n (x_i^3 + x_i \sin x_i) \Delta x = \int_0^{\pi} (x^3 + x \sin x) dx.$$

Example. Evaluate $\int_0^3 (x^3 - 6x) dx$.

Sln. $f(x) = x^3 - 6x$; $\Delta x = \frac{b-a}{n} = \frac{3-0}{n} = \frac{3}{n}$.

$x_i = a + i\Delta x = 0 + \frac{3i}{n} = \frac{3i}{n}$ then.

$$\int_0^3 (x^3 - 6x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x.$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n f\left(\frac{3i}{n}\right) \frac{3}{n}$$

$$= \lim_{n \rightarrow \infty} \frac{3}{n} \sum_{i=1}^n \left(\left(\frac{3i}{n}\right)^3 - 6\left(\frac{3i}{n}\right) \right)$$

$$= \lim_{n \rightarrow \infty} \frac{3}{n} \sum_{i=1}^n \left(\frac{27 i^3}{n^3} - \frac{18i}{n} \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{3}{n} \sum_{i=1}^n \frac{27 i^3}{n^3} - \frac{3}{n} \sum_{i=1}^n \frac{18i}{n} \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{81}{n^4} \sum_{i=1}^n i^3 - \frac{54}{n^2} \sum_{i=1}^n i \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{81}{n^4} \left(\frac{n(n+1)}{2} \right)^2 - \frac{54}{n^2} \cdot \frac{n(n+1)}{2} \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{81}{n^4} \frac{n^2(n+1)^2}{4} - \frac{54}{n^2} \cdot \frac{n^2(1+1/n)}{2} \right)$$

$$= \lim_{n \rightarrow \infty} \left\{ \frac{81}{n^4} \cdot \frac{n^2 \cdot n^2 (1+1/n)^2}{4} - \frac{54}{2} \cdot (1+1/n) \right\}$$

$$= \lim_{n \rightarrow \infty} \left\{ \frac{81}{4} \cdot \underbrace{(1+1/n)^2}_1 - \frac{54}{2} \underbrace{(1+1/n)}_1 \right\} \quad \frac{\#}{\infty} \rightarrow 0$$

$$= \frac{81}{4} - \frac{54}{2} = \frac{81}{4} - 27 = -6.75.$$

$\int_0^3 (x^3 - 6x) dx = F(3) - F(0)$ if F is an antiderivative.

$F(x) = \frac{x^4}{4} - 3x^2$. Thus.

$$\int_0^3 (x^3 - 6x) dx = F(3) - F(0) = \frac{3^4}{4} - 0 = \frac{81}{4}.$$

The midpoint rule:

$$\int_a^b f(x) dx \approx \sum_{i=1}^n f(\bar{x}_i) \Delta x \quad \text{where } \Delta x = \frac{b-a}{n}$$

$$\begin{array}{c} \text{-----} \\ | \quad | \quad | \\ x_{i-1} \quad \bar{x}_i \quad x_i \end{array}$$

$$\bar{x}_i = \frac{x_{i-1} + x_i}{2}$$

Properties of the Definite integral.

i). $\int_a^b f(x) dx = - \int_b^a f(x) dx.$

ii). $\int_a^a f(x) dx = 0$

iii). $\int_a^b c \cdot f(x) dx = c \cdot \int_a^b f(x) dx.$

iv). $\int_a^b (f(x) \pm g(x)) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx.$

Example. Compute $\int_0^1 (4 + 3x^2) dx.$

Sln. $\int_0^1 (4 + 3x^2) dx = \int_0^1 4 dx + \int_0^1 3x^2 dx.$

$$= 4 \int_0^1 1 dx + 3 \int_0^1 x^2 dx.$$

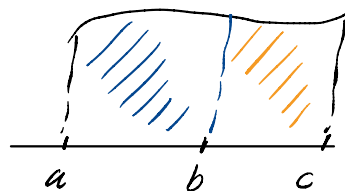
$$= 4x \Big|_0^1 + 3 \cdot \frac{x^3}{3} \Big|_0^1.$$

$$= 4x \Big|_0^1 + x^3 \Big|_0^1$$

$$= (4 - 0) + (1 - 0) = 5.$$

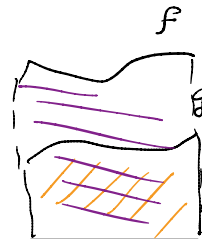
More Properties.

v). $\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$



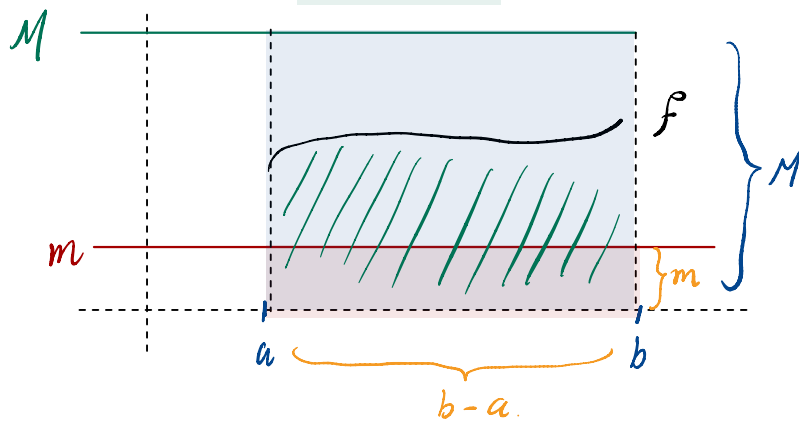
vi). If $f(x) \geq 0$ $a \leq x \leq b$ then $\int_a^b f(x) dx \geq 0.$

vii). If $f(x) \geq g(x)$ for $a \leq x \leq b$ $\int_a^b f(x) dx \geq \int_a^b g(x) dx.$



viii). If $m \leq f(x) \leq M$ for $a \leq x \leq b$ then

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a).$$



How we can compute antiderivatives?

Goal: Evaluate $\int_a^b f(x) dx$ using FTC2.

Indefinite Integrals.

We need a convenient notation for antiderivatives that makes them easy to work with.

$$\int f(x) dx = F(x) \quad \text{mean} \quad F'(x) = f(x).$$

The notation $\int f(x) dx$ is traditionally used for an antiderivative of f and is called an indefinite integral.

$$\begin{aligned} \int_a^b f(x) dx &\longrightarrow \text{Area under the curve } y=f(x). \\ &\quad \text{but.} \\ \int f(x) dx &\longrightarrow \text{Antiderivative of } f(x). \end{aligned}$$

For example: $\int x^2 dx = \frac{x^3}{3} + C.$
 b/c $\frac{d}{dx} \left(\frac{x^3}{3} + C \right) = x^2.$

Remk: You should distinguish carefully between definite and indefinite integrals. A definite integral $\int_a^b f(x) dx$ is a number and an

indefinite integral $\int f(x) dx$ is a function (or family of functions).

The connection between them is given by.

$$\int_a^b f(x) dx = \underbrace{\int f(x) dx}_\text{an antiderivative} \Big|_a^b$$

Example. $\int \sec^2 x dx = \tan x + C$ b/c $\frac{d}{dx} (\tan x + C) = \sec^2 x$.

Table of indefinite integrals.

1). $\int c f(x) dx = c \int f(x) dx$.

2). $\int k dx = kx + C$.

3). $\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$.

4). $\int x^n dx = \frac{x^{n+1}}{n+1} + C$.

5). $\int \sin x dx = -\cos x + C$.

6). $\int \sec^2 x dx = \tan x + C$.

7). $\int \cos x dx = \sin x + C$.

8). $\int \csc^2 x dx = -\cot x + C$.

9). $\int \sec x \tan x dx = \sec x + C$.

10). $\int \csc x \cot x dx = -\csc x + C$.

Example. $\int \frac{1}{x^2} dx = \int x^{-2} dx = \frac{x^{-2+1}}{-2+1} + C = -\frac{1}{x} + C$.

Example. Find

i). $\int (10x^4 - 2\sec^2 x) dx$.

ii). $\int \frac{\cos \theta}{\sin^2 \theta} d\theta$.

Soln. i). $\int (10x^4 - 2\sec^2 x) dx = \underbrace{10 \int x^4 dx}_{\text{using 4.}} - 2 \underbrace{\int \sec^2 x dx}_{\text{using 6.}}$

$$= 10 \frac{x^{4+1}}{4+1} - 2 \cdot \tan x + C.$$

$$= 2x^5 - 2 \tan x + C.$$

$$\frac{d}{dx} (2x^5 - 2 \tan x + C) = 10x^4 - 2 \sec^2 x.$$

$$ii). \int \frac{\cos \theta}{\sin^2 \theta} d\theta = \int \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\sin \theta} d\theta = \int \underbrace{\cot \theta}_{\text{using 10.}} \cdot \csc \theta d\theta = -\csc \theta + C.$$

Example. Find $\int_0^{12} (x - 12 \sin x) dx$.

Sln. $\int (x - 12 \sin x) dx = \int x dx - 12 \int \sin x dx$

$$= \frac{x^2}{2} - 12(-\cos x) + C.$$

$$= \frac{x^2}{2} + 12 \cos x + C.$$

$$\int_0^{12} (x - 12 \sin x) dx = \left(\frac{x^2}{2} + 12 \cos x + C \right) \Big|_0^{12}$$

$$= \frac{12^2}{2} + 12 \cos 12 + C - \left(\frac{0^2}{2} + 12 \cos 0 + C \right)$$

$$= 72 + 12 \cos 12 + \cancel{C} - 12 - \cancel{C}$$

$$= 60 + 12 \cos 12.$$

Problem: Evaluate $\int_1^9 \frac{2t^2 + t^2 \sqrt{t} - 1}{t^2} dt$.